

PDF AI Assistant Journal

Right off the bat, I have decided to utilize ungated Hugging Face LLMs, since I have significantly more experience with them, and I did not want to incur costs such as if I opted to go with OpenAI API that will require an API key. Given this, I opted to use TinyLlama/TinyLlama-1.1B-Chat-v1.0 lightweight model, also to save runtime since this is only a short app demo, and it is expected that the model will have more instances of hallucination. This is why I worked on integrating RAG, which retrieves the top three most relevant chunks from the uploaded document per query, to maximize reliability and combat potential hallucination.

TinyLlama was fine-tuned with a chat format using `<|system|>`, `<|user|>`, and `<|assistant|>` tags, which I have incorporated in my system prompt to keep consistency and compatibility with the model.

My iterations include changing a separate PDF upload button to one included in the chat window with `st.chat_input`, correcting the prompt template to the TinyLlama format, and adding a chunk retriever limit to the RAG retriever.

I have to admit that I was very ambitious and wanted to explore more directions. I wanted to explore different RAG indexing and query translation techniques, such as ColBERT and RAG-fusion, that would hopefully improve retrieval and generation. Moreover, I wanted to integrate parameter-efficient fine-tuning (PEFT) on top of RAG, which would effectively improve answer reliability and grounding. Lastly, I wanted to make the assistant not exclusive to PDFs alone but also cater to other text document file formats. However, given the very limited time, I was not able to pursue these directions.

I would like to note that this is the first time I worked on Streamlit and I had a lot of fun learning its basics, and would definitely work on other projects with Streamlit in the future.